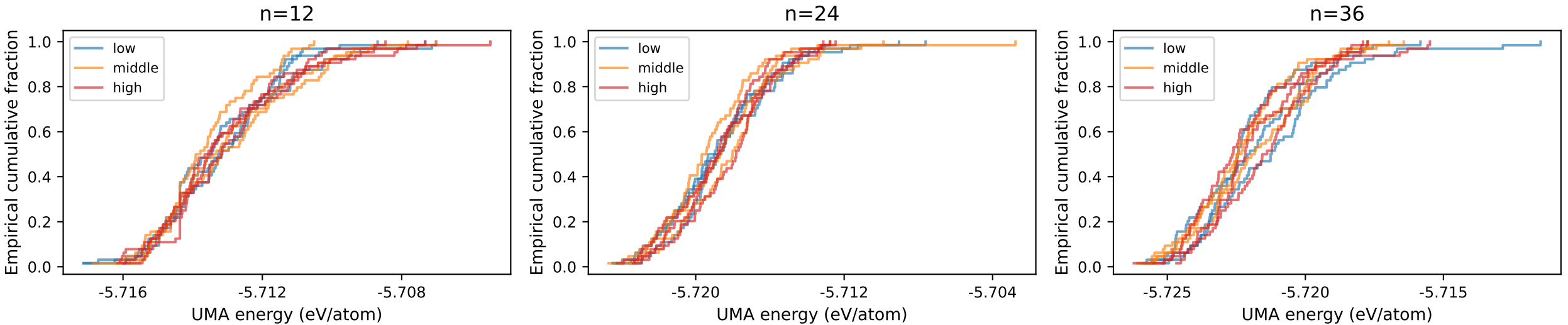dense; thin curves are design replicates